

Intraoperative Hypotension: Definitions, Thresholds, and Incidence

Based on ESA/ASA Consensus Guidelines & Circulation 2021

1. Definition

Intraoperative hypotension (IOH) is most widely defined as a mean arterial pressure (MAP) below 65 mmHg for any duration during a surgical procedure. This threshold is endorsed by the European Society of Anaesthesiology (ESA) and reflects the lower boundary of cerebral autoregulation and renal perfusion pressure in the majority of adult patients.

Alternative definitions that appear in the literature include:

- Absolute: $\text{MAP} < 60$ or < 55 mmHg
- Relative: a decrease of more than 20% or 30% from a pre-defined baseline MAP
- Time-based: $\text{MAP} < 65$ mmHg for a cumulative duration exceeding 1 minute

The $\text{MAP} < 65$ mmHg absolute threshold is preferred in predictive modeling because it is independent of pre-operative variability and maps directly to organ perfusion endpoints.

2. Incidence

IOH is among the most common intraoperative complications, with reported incidence ranging from 5% to 99% depending on the definition used and the surgical population studied.

Using the $\text{MAP} < 65$ mmHg threshold:

- Non-cardiac surgery (general population): 15–30%
- Elderly patients (≥ 65 years): 30–55%
- Major abdominal surgery: 25–40%
- Spinal anesthesia cases: 30–60%

The VitalDB dataset used by this system contains 6,388 surgical cases from Seoul National University Hospital and shows an overall IOH rate of approximately 10–15% using the 1-minute sustained threshold employed by the ML prediction model.

3. Duration Matters

The relationship between IOH severity and organ injury is non-linear. Even brief episodes matter:

- MAP 55–64 mmHg for ≥ 1 min: associated with increased AKI risk (OR 1.4)
- MAP < 55 mmHg for ≥ 5 min: associated with myocardial injury (OR 2.1)
- MAP < 65 mmHg for ≥ 11 min cumulative: associated with 30-day mortality (HR 1.3)

These thresholds are from the POISE-3 trial and Wesselink et al., Anesthesiology 2018.

The VitalWatch alert system flags MAP trajectories before they breach these thresholds.